messages. It can change the state of the server, and the device having the server model will respond by switching on or off.

Light lightness server, or LLS, controls brightness and dimming of lights, while light hue saturation lightness server, or LHSLS, is for colour control. Light lighting control server, or LLCS, allows interfacing of sensors with light(s) to accomplish useful effects.

State bindings are relationships between state items. If a particular state changes its value through state binding, then a change can automatically be triggered in another state inside the device in another model, perhaps, for which this special relationship has been specified.

Now, imagine the occupancy sensor detecting people entering and leaving a room. When you walk into the room, the sensor is sampling occupancy data and will send a message using the sensor model to LLCS to indicate occupancy. LLCS gets to know about this occupancy by registering its value in the state server. Thanks to state binding server for making a connection between LLCS and generic on/off server, changing occupancy state will automatically change on/off state. In other words, as you walk into the room, the sensor will notice your presence and the light will switch on. We can build relationships between these models to achieve all sorts of useful things.

In place of an occupancy sensor, imagine having an ambient light sensor. This is a common requirement in smart buildings. With the ambient light sensor, light levels can be sampled and as changes occur, messages can be sent to report ambient light levels to LLCS model. It has a state binding with LLS. When the sensor detects brightness from an external source (bright sunlight), then the light will get dim or may get completely turned off.

Conversely, when it gets darker, the light will get brighter. A time server can also be included in the model to schedule time changes.

Smart lights consist of some hardware, possibly some sensors integrated into that same unit. These include a microcontroller (MCU) and Bluetooth Mesh communications component that can communicate with other devices in the mesh network. At the top of this stack, there is application software that controls the lights (switching on/off, changing colours, etc). For multiple lights, multiple applications can be implemented on that same MCU. This way every single light can become a beacon.

Bluetooth beacons are becoming increasingly common. These are Bluetooth transmitters that broadcast a unique ID. If you are walking along a smartphone with some software on it, and it receives that broadcast with that unique ID, it can translate that ID into knowledge of a place or an object. Bluetooth beacons are often used to give localised information when you are in proximity of something, or to give indoor navigation capabilities. So, in a really large building like an airport, you can deploy an indoor navigation system because you have the infrastructure.

Ways of creating a mesh network

There are two distinct ways of creating a mesh network. One involves routing, which is figuring out how to get data from point A to B via intermediate points. There are algorithms that determine routes. However, routing gives rise to a single point of failure in the network, and that is not acceptable. Routing is an efficient way of handling data, but it is not sufficiently reliable.

Flooding, on the other hand, is at the other end of the spectrum and involves no routing whatsoever. A message gets broadcast, and every device that receives that broadcast broadcasts it again. So, the message spreads across the whole network and everything receives everything. It gets through in a reliable way, given the amount of re-transmission that goes on. But it is not very efficient.

Bluetooth Mesh uses something between the two, called managed flooding, which uses some fundamentals of flooding—routing is not done to avoid single points of failure. But there are various strategies of doing flooding that make it efficient. This gives the best of both the worlds. You can control how many hops a message will do. There are heartbeat messages flowing across the network, so devices can inform each other about where they are or how far they are. This way the